

Feedback

by Diego Navarra- Friday, 12 December 2025, 5:48 PM

1. Knowledge and Understanding

This literature review demonstrates a strong and well-articulated understanding of the intersection between Internet of Things (IoT) technologies and sustainable water management in the UAE's arid environment. The author situates the topic effectively within national sustainability agendas such as Smart Dubai 2021 and Abu Dhabi Vision 2030, linking these to the broader discourse on digital transformation and smart utilities. The description of IoT's three-layer architecture (sensing, network, and application) is clear, technically sound, and shows strong conceptual grasp.

The review also illustrates how IoT technologies—ranging from smart metering to predictive maintenance—address specific challenges of water scarcity, desalination dependency, and inefficient irrigation. The contextual alignment between technology, climate conditions, and infrastructure constraints in the UAE demonstrates very good applied understanding. This work meets postgraduate-level expectations for both technical comprehension and contextual relevance.

2. Criticality

The review contains elements of critical reflection, particularly when discussing implementation barriers such as cost, data security, and workforce capability. The acknowledgement of discrepancies in reported cost-benefit analyses and integration challenges indicates awareness of research limitations and conflicting findings.

However, while the review is analytically structured, it tends to remain more descriptive than evaluative in tone—summarising studies rather than interrogating their methodologies, assumptions, or generalisability. The discussion would benefit from stronger synthesis and comparative evaluation, such as exploring why certain pilot studies (e.g., UAE vs. Saudi Arabia or Kuwait) yield differing performance outcomes, or assessing the reliability of small-scale trials. A more explicit critical framework—linking evidence to concepts like scalability, sustainability trade-offs, or governance maturity—would strengthen the analytical dimension.

3. Use of Relevant Sources

The literature base is robust, current, and relevant, spanning peer-reviewed journal articles, conference papers, and technical reports from 2015–2024. The inclusion of recent works (e.g., Et-taibi et al., 2024; Hassan et al., 2024; Alshami et al., 2023) shows an awareness of the latest technological developments. The references are appropriately multidisciplinary, covering engineering, environmental management, and policy.

All sources are properly cited using Harvard referencing, and the reference list is accurate and complete. Integration of sources into the argument is clear and consistent. To achieve higher distinction-level performance, the review could incorporate more comparative analysis of global frameworks or case studies, connecting international evidence with UAE-specific conditions to build a stronger critical argument.

4. Structure and Presentation

The structure follows an exemplary and professional format, progressing logically from introduction through conceptual foundations, applications, and future research directions. The subheadings are well chosen and provide clarity for the reader. The narrative flow is coherent, and transitions between sections are well managed.

The writing style is clear, formal, and academic. Paragraphs are well developed, though some could be more concise to avoid repetition of concepts such as “real-time monitoring” and “predictive maintenance.” Presentation quality—including formatting, layout, and referencing style—is excellent and consistent with postgraduate academic standards.

5. Academic Integrity

Academic integrity is of a very high standard. All materials are correctly attributed and referenced, with no evidence of plagiarism or inappropriate citation practice. The author demonstrates responsible and ethical engagement with source material, maintaining originality in synthesis and phrasing.